Calculations with the standard and general normal distributions

1 If $X \sim N\left(20, 5^2\right)$, use your calculator to find the value of k in the following parts:

a $P(X < k) = 0.65$

b $P(X > k) = 0.45$

c $P(k < X < 27) = 0.89$

d $P(21 < X < k) = 0.4$

2 Languages and Mathematics are very different disciplines, and so to compare results in the two subjects, the standard deviation is used. The mean and standard deviation of exam results in each subject are given in the table:

	Mean	Std. Deviation
Languages	60	7
Mathematics	67	8

a A student receives a mark of 81 in Languages. How many standard deviations away from the mean is this mark?

b What mark in Mathematics would be equivalent to a mark of 81 in Languages?

c A student receives a mark of 86.2 in Mathematics. How many standard deviations away from the mean is this mark? Round your answer to one decimal place.

d What mark in Languages would be equivalent to a mark of 86.2 in Mathematics? Round your answer to one decimal place.

Quantiles and percentiles

3 For the standard normal variable $X \sim N(0, 1)$, use a CAS to determine the following values to three decimal places:

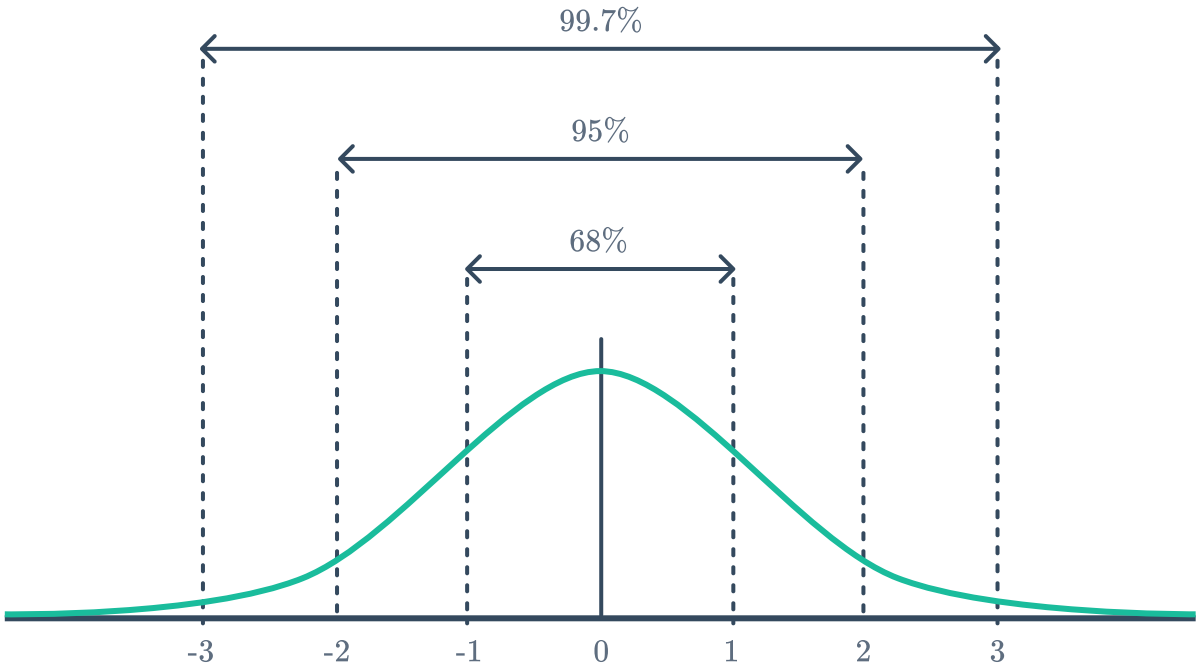
a The 0.7 quantile

b The 65th percentile

c The lowest score in the top 20 percent

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4 Consider the graph of a standard normal distribution showing the $68-95-99.7$ rule:



a Which value is the closest to the 0.5 quantile?

b Which value is the closest to the 0.84 quantile?

c Which value is the closest to the 16th percentile?

5 Consider a normal distribution defined by $X \sim N(50, 25)$. Use the $68-95-99.7$ rule to answer the following questions:

a Which value is equivalent to the 0.16 quantile?

b Which value is equivalent to the 0.025 quantile?

c Which value is equivalent to the 97.5th percentile?

6 The heights of a certain species of fully grown plants are thought to be normally distributed with a mean of 40 cm and a standard deviation of 1 cm. Use the $68-95-99.7$ rule to answer the following questions:

a What is the height of the shortest plant in the 84th percentile?

b What is the height of the shortest plant in the 0.0015 quantile?

7 For a normal variable defined by $X \sim N(100, 100)$, use a CAS to determine the following values to three decimal places:

a The 0.2 quantile

b The 90th percentile

c The lowest score that is greater than the bottom 30 percent

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8 If $X \sim N\left(30, 4^2\right)$, calculate:

a The 0.5 quantile

b The 0.83 quantile

c The 35th percentile

9 A random variable is normally distributed such that $X \sim N\left(50, 25\right)$.

a Calculate the standard score if $X = 58$.

b Using a calculator or otherwise, calculate the z-score for the 0.35 quantile.

c Hence, find the X value for the 0.35 quantile.

10 Mensa is an organisation that only accepts members who score in the 98th percentile or above in an IQ test. Explain what a person has to do to get into Mensa.

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